

What is the need for studying Next Generation Databases in ADBMS session

Next-generation database refer to advanced database systems developed after traditional relational databases to handle **modern data requirements**.

Examples include:

- Object-Oriented Databases (OODBMS)
- NoSQL Databases
- Distributed Databases

So Traditional relational databases (RDBMS) are powerful, but they have limitations in today's data-driven world.

1. Handling Complex Data

- Modern applications deal with:
Images ,Videos ,Audio , Graph data

RDBMS cannot efficiently store and process such complex data

2. Big Data & Large Volume

Today's systems generate: Huge volumes of data (TBs, PBs)

Next-gen DBs provide: Scalability Distributed storage

3. High-Speed Processing

Real-time applications require: Fast data access , Low latency

Example : Online banking Stock market systems

4. Flexibility in Data Model

RDBMS requires: Fixed schema

Next-gen DBs allow:

Dynamic schema (schema-less)

Useful in: Social media apps E-commerce platforms

5. Distributed Environment Data is stored across multiple locations

Next-gen DBs support: Distributed processing Cloud-based systems

6. Object-Oriented Application Support

Modern programming uses: Java, Python (object-based)
Need databases that support: Objects directly (OODBMS)

7. Reduced Complexity in Development

No need for: Complex joins
Object-relational mapping
Faster application development

8. High Availability & Fault Tolerance

Systems must work **24/7** Next-gen DBs provide: Replication Automatic recovery

9. Support for Modern Applications

Used in:

AI & Machine Learning
IoT (Internet of Things)
Big Data analytics
Cloud computing